List operations:

ListOperations.py

```python
numbers=[10,20,40]
numbers.insert( index: 2, object: 30)
print(numbers)
numbers=[10,20,30,40]
numbers.remove(30)
print(numbers)
numbers.pop()
print(numbers)
print(len(numbers))
numbers=[10,20,30,40]
for num in numbers:
    print(num)
fruits=["apple","banana","mango"]
if "banana" in fruits:
    print("Banana exists")
numbers=[10,20,30,40,50]
print(numbers[1:4])
numbers=[10,20,30,40]
numbers.reverse()
print(numbers)
numbers.sort()
print(numbers)
print(max(numbers))
print(min(numbers))
print(sum(numbers))
```

Tuple Operations:

```
numbers=(10,20,30,40)
print(numbers)
fruits=("apple","banana","mango")
print(fruits[0])
print(fruits[2])
fruits=("apple","banana","mango")
print(fruits[-1])
print(fruits[-2])
numbers=(10,20,30,40,50)
print(len(numbers))
numbers=(10,20,30,40)
for n in numbers:
    print(n)
numbers=(10,20,30)
numbers[1]=250
```

Packing and unpacking tuple operations:

```
student=("Rahul",22,"Python")
print(student)
name,age,course=student
print(name)
print(age)
print(course)
data=("Arjun",25,True,5000.50)
print(data)
```

Set operations:

setoperations.py

```python
numbers={10,20,30}
print(numbers)
numbers={10,20,30,40,50,50}
print(numbers)
numbers=[10,20,30,40,50,50]
unique_numbers=set(numbers)
print(unique_numbers)
numbers={10,20,30}
numbers.add(40)
print(numbers)
numbers={10,20,30}
numbers.update([40,50,60])
print(numbers)
set1={10,20,30}
set2={30,40,50}
result=set1.union(set2)
print(result)
result1=set1.difference(set2)
print(result1)
result2=set1.intersection(set2)
```

Dictionary operations

dictionary.py

```python
student={
    "name":"Rahul",
    "age":25,
    "course":"python"
}
print(student)
print(student["name"])
print(student["age"])
print(student["course"])
print(student.get("name"))
print(student.get("age"))
print(student.get("course"))
student["city"]="Hyderabad"
print(student)
```

FILE HANDLING:

Data file with read mode:

```
Rahul
priya
anadha
ragav
```

```python
with open("data.txt","r") as file:
    for line in file:
        print(line.strip())
with open("data.txt","r") as file:
    students=file.readlines()
print("Total students:",len(students))
```

mathdata.txt

```
10
20
30
40
50
```

mathdata.txt | numfilehandling.py

```python
total=0
with open("mathdata.txt","r") as file:
    for line in file:
        total=total+int(line)
print("Total:",total)
```

Write mode:

writeoperations.py data.txt

```python
with open("data.txt","w") as file:
    file.write("Rahul\n")
    file.write("Priya\n")
    file.write("Arjun\n")
```

Appending data:

```python
with open("data.txt","a") as file:
    file.write("Nithish\n")
```

Writelines method:

```python
languages=["python\n","java\n","c\n"]
with open("data.txt","w") as file:
    file.writelines(languages)
```

Working with json:

writeoperations.py {} data.json

```json
{
  "students":[
  {"name": "Rahul","marks": 85},
  {"name": "Sneha","marks": 92},
  {"name": "Arjun","marks": 78}
]
}
```

```python
import json
with open("data.json","r") as file:
    data = json.load(file)
print(data)
for student in data["students"]:
    print(student["name"],student["marks"])

students={
    "students":[
        {"name":"Priya","marks":88},
        {"name":"Karnan","marks":78}
    ]
}
with open("output.json","w") as file:
    json.dump(students,file)
```

Indentation:

```python
students={
    "students":[
        {"name":"Priya","marks":88},
        {"name":"Karnan","marks":78}
    ]
}
with open("output.json","w") as file:
    json.dump(students,file,indent=4)
```

Working with csv files:

data.csv

```csv
name,marks
Rahul,85
Sneha,72
Arjun,98
```

```python
import csv
with open("data.csv","r") as file:
    reader = csv.reader(file)
    for row in reader:
        print(row)
with open("data.csv","r") as file:
    reader=csv.DictReader(file)
    for row in reader:
        print(row["name"],row["mark"])
```

```python
with open("output.csv","w",newline="") as file:
    writer=csv.writer(file)
    writer.writerows(data)
```